

dialog box will open; select Projects followed by DataTransferObjectLib from it. Click OK.

8. Now, go back to our AttendanceRegistrationSystem project and add references to the two projects created so far as shown in Fig. 18.

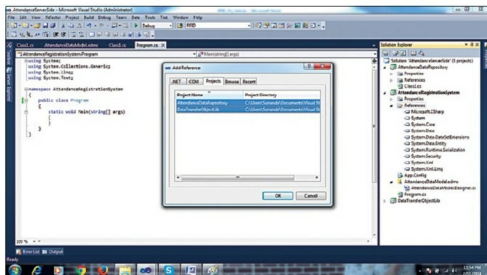


Fig. 18: Adding reference

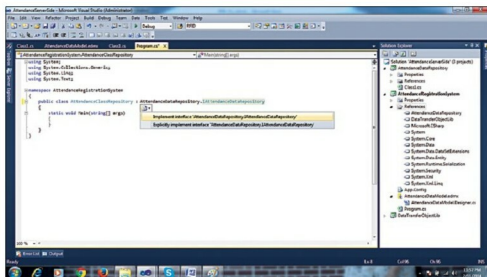


Fig. 19: AttendanceDataRepository

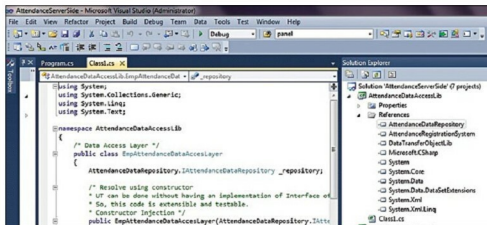


Fig. 20: EmpAttendanceDataAccessLayer

9. In this step, rename the class in Program.cs from Program to AttendanceClassRepository and then implement our IAttendanceDataRepository interface (refer Fig. 19). All its abstract methods get implemented in this class.

10. Create CreateEmpDetails

method. For this, we create an instance out of EDM runtime, that is, EmployeeAttendanceDBEntities. Then, we construct DTOs based on user data sent by higher layers. Finally, we add objects to their respective Data Objects in runtime and then save the changes.

11. Similarly, we write the logic to retrieve employee details based on EmpId and RFID, and return a list or an object, respectively.

**Note.** Here, we will use the concept of Language Integrated Query (LINQ) to query the database. LINQ is a set of features introduced in Visual Studio 2008 that extends powerful query capabilities to the language syntax of C# and Visual Basic.

12. To update the data in table CurrEmpDetail, the data object passed by the user is taken and CurrEmpDetail data object of EmployeeAttendanceDBEntities EDM runtime is updated accordingly and saved into the database.

The interaction part with the database is now complete. We will now write actual data-access library for this application.

13. Create a class library AttendanceDataAccessLib into AttendanceServerSide solution. Rename the public class inside Class1.cs to EmpAttendanceDataAccessLayer (Fig. 20). Now, add references of DataTransferObjectLib, AttendanceRegistrationSystem and AttendanceDataRepository to the project.

Inside the constructor of EmpAttendanceDataAccessLayer, we instantiate our repository class (if not set already) and create methods to push and retrieve data from the repository.

To be concluded next month

## EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at [source.efymag.com](http://source.efymag.com)



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